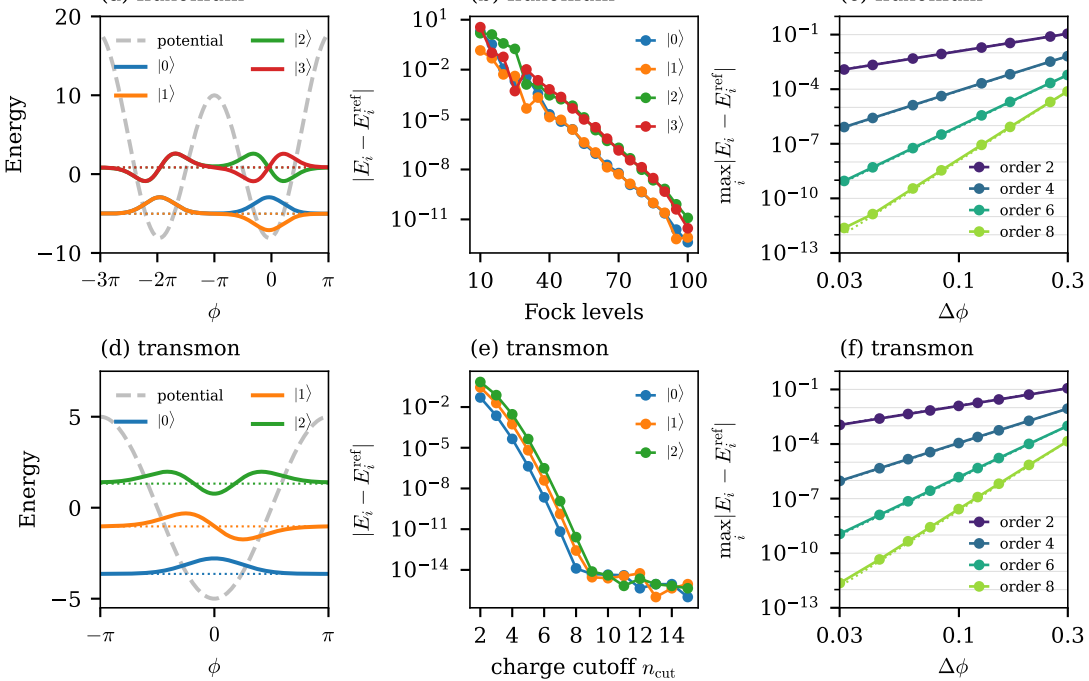
(a) fluxonium
potential
|0⟩
|1⟩
|2⟩
|3⟩
Energy
20
10
0
−10
−3π
−2π
−π
0
π
ϕ
(b) fluxonium
$|E_i - E_i^{\mathrm{ref}}|$
10^{1}
10^{-2}
10^{-5}
10^{-8}
10^{-11}
10
40
70
100
Fock levels
(c) fluxonium
$\max_i |E_i - E_i^{\mathrm{ref}}|$
10^{-1}
10^{-4}
10^{-7}
10^{-10}
10^{-13}
order 2
order 4
order 6
order 8
0.03
0.1
0.3
Δϕ
(d) transmon
potential
|0⟩
|1⟩
|2⟩
Energy
5
0
−5
−π
0
π
ϕ
(e) transmon
$|E_i - E_i^{\mathrm{ref}}|$
10^{-2}
10^{-5}
10^{-8}
10^{-11}
10^{-14}
2
4
6
8
10
12
14
charge cutoff n_{cut}
(f) transmon
$\max_i |E_i - E_i^{\mathrm{ref}}|$
10^{-1}
10^{-4}
10^{-7}
10^{-10}
10^{-13}
order 2
order 4
order 6
order 8
0.03
0.1
0.3
Δϕ